

Day-1: Introduction to Python

What is a program?

- → A program is the process of instructions given by the user or programmer to plan what to do and how to do...

Ex:

```
user_en = 2000
bal_ = 1000
result = bal_ - user_en
print(result)
```

What is Procedural Language?

→ A Procedural language is organized mainly around functions

Ex:

C-language

```
def cal_ad(a,b):
    return a+b

a = 7
b = 9
any_ = cal_ad(a,b)
print(any_)
```

What is Object-Oriented programming?

→ Object-Oriented programming is organized mainly on classes and objects.

Ex:

```
num = 7
print(type(num))
```

What is Python?

→ High-level language

→ Python takes care of memory allocation

Ex:

```
num = 9
print(type(num))
print(id(num))
```

→ Interpreter language

→ Line-by-line execution is followed by Python; that is the reason we call it an interpreter language

Ex:

```
num = 9
num_2 = 90
print(num)
print(type(num))
print(id(num))
```

→ Dynamically typed language

→ No need to mention the type of data passed to the variable

Ex:

```
num =78
```

```
any_ ='python'
```

```
nums = [1,2]
```

```
all_= (1,2)
```

```
print(type(num))
```

```
print(type(any_))
```

```
print(type(nums))
```

```
print(type(all))
```

Why Python?

1. More library
2. Cross -platform
3. Open-Source
4. Simple Syntax

Applications

1. Use in Data science
2. Web Development
3. AI
4. ML
5. DL

History of Python

→It was started in the early 1980s and released in 1991 by Guido Van Rossum, and he picked the name from the most loved series called "Monty python's"

→ The first version was released in the year 1991, which is Python 0.9.0

→ The latest version we are using is Python 3.14